David Elliott Perryman

website | David Elliott Perryman | elliottperryman | elliott.perryman@hey.com

Work

PhD Candidate at Institute Laue-Langevin Mar 2024 - Sept 2027

- Developing autonomous experimentation methods for neutron scattering experiments on quantum materials.
- Bringing modern Bayesian stats (Hamiltonian Monte-Carlo / variational inference) and automatic differentiation to physics problems.
- Creating provably d-optimal design plans for neutron reflectivity experiments.
- Creating reinforcement learning approaches to sequential Bayesian Optimal Experiment Design problems
- Communicated results to technical and non-technical audiences and built instructional demos
- Elected to represent the interests of 40+ PhD students to management, unions, and the French labor inspectorate as well as organize community events.

Software Developer at Caravel Concepts (Remote) Jan 2021 - Aug 2021

- Connected bank API endpoints to PostreSQL database via django
- Improved quality of asset allocation optimization with JAX, Numba, and linear algebra.
- Collaborated across a cross-functional team to produce software

Undergraduate Researcher at Lawrence Berkeley Laboratory Aug 2019 - Jun 2020

- 20x accelerated ALS-steering software GPCAM through batching, converting Cholesky decomposition to linear solves, and porting to GPU with PyTorch.
- Created fast numerical integration methods in Python for crystallography.
- Developed a hybrid, distributed optimization method (HGDL).
- Communicated technical results through conference papers, talks, and articles

Undergraduate Researcher at Oak Ridge National Labs Aug 2017 - Jun 2019

- Ported signal processing convolution to CUDA FFT on GPU for 30X speedup
- Characterized and optimized signal feature extraction using Monte-Carlo methods
- Wrote signal anomaly detection with Conv Nets and data exploration with k-Means

Education

PhD (Physics) at **Université Grenoble Alpes** Mar 2024 - present
Master's (Applied Math) at **Institut Polytechnique de Grenoble** Aug 2021 - May 2023
Bachelor's (Comp Sci & Physics) at **UT Knoxville** Sept 2016 - May 2021

Languages	C, C++, CUDA, Julia, Python, R, SQL, English, French (A2)
Python Tools	Jax, Numba, numpyro, Pandas, scikit-learn, matplotlib, cvxpy
Interests	Applied Bayesian Statistics, Applied Math, Physics, Experimental Design
Methods	Variational Inference, Hierarchical Bayesian Models, Linear Programming, Convex Optimization, Gaussian Processes, Fourier Analysis
Work Authorization	US (citizen) and France (resident).